Kuiper found, methane, $CH_4$, the starting material from which carbon-based organic molecules are generated there.

A variety of simple organic molecules was found, present as gases, mainly hydrocarbons and nitriles. The most complex of them have four "heavy" (carbon and/or nitrogen) atoms. Hydrocarbons are molecules composed of carbon and hydrogen atoms only, and are familiar to us as natural gas, petroleum, and waxes. (They're quite different from carbohydrates, such as sugars and starch, which also have oxygen atoms.) Nitriles are molecules with a carbon and nitrogen atom attached in a particular way. The best known nitrile is HCN, hydrogen cyanide, a deadly gas for humans. But hydrogen cyanide is implicated in the steps that on Earth led to the origin of life.

Finding these simple organic molecules in Titan's upper atmosphere—even if present only in a part per million or a part per billion—is tantalizing. Could the atmosphere of the primeval Earth have been similar? There's about ten times more air on Titan than there is on Earth today, but the early Earth may well have had a denser atmosphere.

Moreover, *Voyager* discovered an extensive region of energetic electrons and protons surrounding Saturn, trapped by the planet's magnetic field. During the course of its orbital motion around Saturn, Titan bobs in and out of this magnetosphere. Beams of electrons (plus solar ultraviolet light) fall on the upper air of Titan, just as charged particles (plus solar ultraviolet light) were intercepted by the atmosphere of the primitive Earth.

So it's a natural thought to irradiate the appropriate mixture of nitrogen and methane with ultraviolet light or electrons at very low pressures, and find out what more complex molecules can be made. Can we simulate what's going on in Titan's high atmosphere? In our laboratory at Cornell—with my colleague W. Reid Thompson playing a key role—we've replicated some of Titan's manufacture of organic gases. The simplest hydrocarbons on Titan are manufactured by ultraviolet light from the Sun. But for all the other gas products, those made most readily by electrons in the laboratory